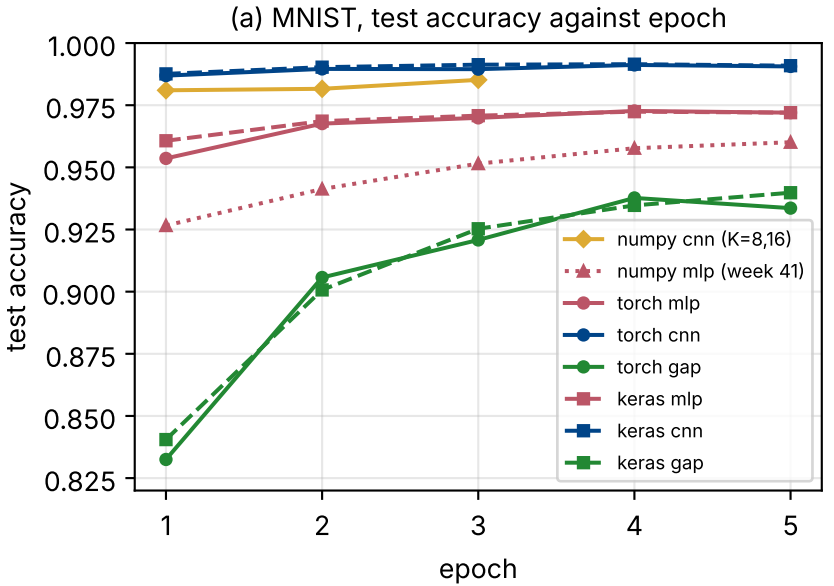
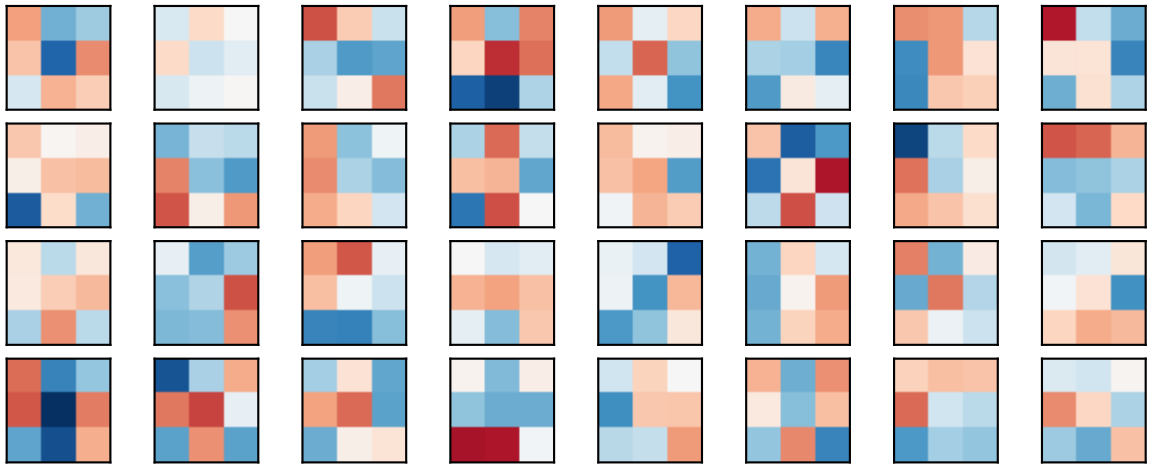
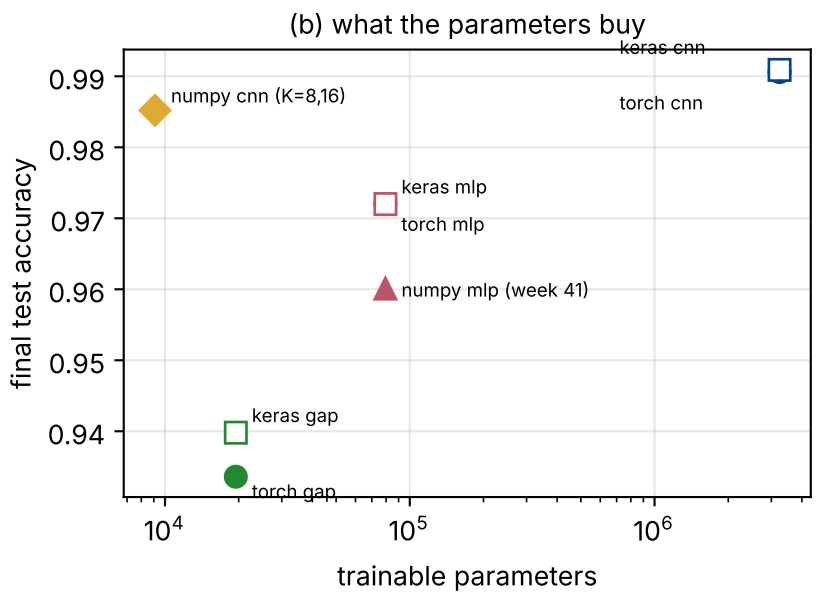
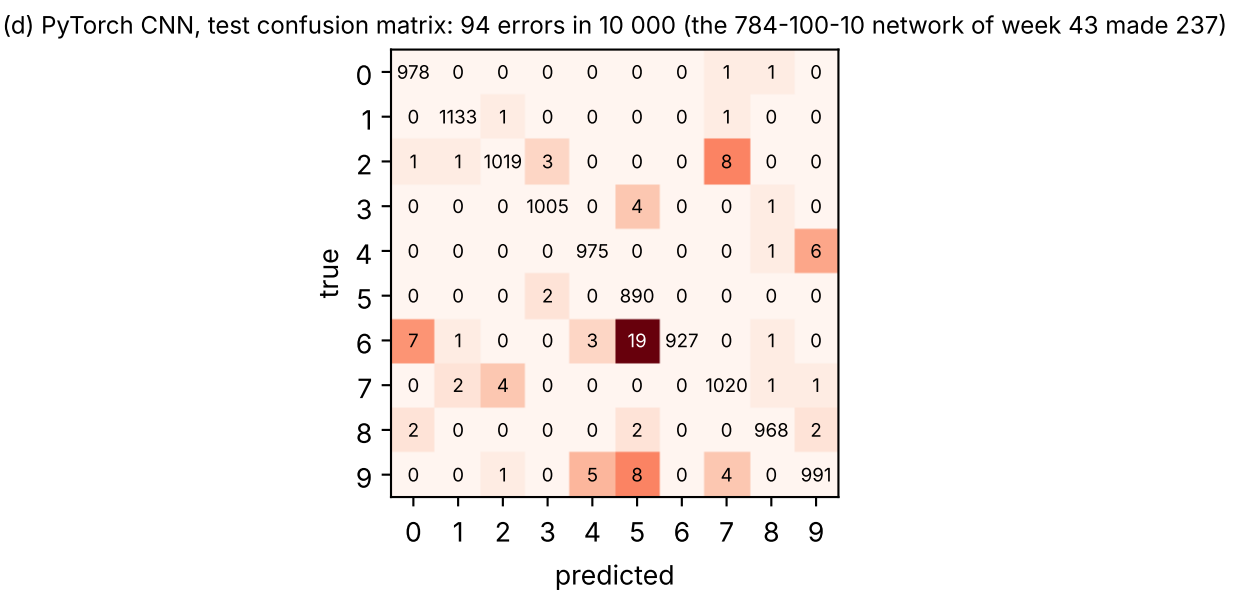

(a) MNIST, test accuracy against epoch
test accuracy
epoch
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.825
1
2
3
4
5
numpy cnn (K=8,16)
numpy mlp (week 41)
torch mlp
torch cnn
torch gap
keras mlp
keras cnn
keras gap


(c) the 32 first-layer 3 × 3 filters of the PyTorch network after training


(b) what the parameters buy
final test accuracy
trainable parameters
0.99
0.98
0.97
0.96
0.95
0.94
$10^4$
$10^5$
$10^6$
keras cnn
torch cnn
numpy cnn (K=8,16)
keras mlp
torch mlp
numpy mlp (week 41)
keras gap
torch gap


(d) PyTorch CNN, test confusion matrix: 94 errors in 10 000 (the 784-100-10 network of week 43 made 237)
true
predicted
0 978 0 0 0 0 0 0 1 1 0
1 0 1133 1 0 0 0 0 1 0 0
2 1 1 1019 3 0 0 0 8 0 0
3 0 0 0 1005 0 4 0 0 1 0
4 0 0 0 0 975 0 0 0 1 6
5 0 0 0 2 0 890 0 0 0 0
6 7 1 0 0 3 19 927 0 1 0
7 0 2 4 0 0 0 0 1020 1 1
8 2 0 0 0 0 2 0 0 968 2
9 0 0 1 0 5 8 0 4 0 991
0 1 2 3 4 5 6 7 8 9